

Falling short: on India's EV journey

India must ensure technology transfer in the EV segment

On June 2, India **took a turn** for the better in its transport electrification journey by offering a **concessional import duty** of 15% on completely built-up units. This is **contingent on** the EV manufacturer investing a minimum of about ₹4,150 crore over three years to **localise** manufacturing in India, with a base domestic value add of 25% in three years, going up to 50% in another two years. The **notification**, under the Scheme to Promote Manufacturing of Electric Passenger Cars in India (SPMEPCI) announced in March 2024, **allows** for a maximum import of 8,000 completely built units annually for each manufacturer for five years. The SPMEPCI **adds to** the **bouquet** of policies that attempts to boost EV adoption and manufacturing. However, these policies put together **fall short of addressing** a **pressing** issue in India's journey to **decarbonise** and transform **mobility** — technology transfer. India began this journey in 2015, about five years later than most large **economies**. An **outlay** of ₹895 crore for the Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles in India (FAME) scheme, for five years, **expanded** to ₹10,000 crore in 2019. China announced its **ambitious** New Energy Vehicle subsidy programme in 2009, which, **coupled** with mandatory **joint venture** manufacturing of EVs until 2022, enabled technology transfer. In addition, a reduced import **duty** on EVs (25% in 2010 to 15% in 2018), and **cumulative incentives** of about \$230 billion in the past 15 years — the most by any country — **enabled** China to achieve the highest global EV adoption rate. This also supported **rapid** charging **infrastructure deployment**, making China the largest producer and consumer of EVs.

The U.S. began this journey in 2010 with an initial outlay of \$25 billion for its Advanced Technology Vehicles Manufacturing Loans Program. This was greatly expanded under the Biden administration's Inflation Reduction Act. But its EV **adoption rate** is much lower than China's. In 2024, out of 17 million global EV car sales, China alone **accounted for** 11.3 million, followed by Europe with 3.2 million, the U.S. with another 1.5 million, and the rest of the world accounting for the **remainder**. China's **vertical integration** of battery manufacturing, from mining, processing to assembling, **has aided economies of scale** with competitive pricing of EVs against **conventional** ICE vehicles. For now, the 25% DVA that India could aim for under the just announced scheme would be **repurposing** locally made auto components meant for ICE vehicles to EVs and **layering** it with **Software-as-a-service**. But to obtain the crucial technology for the heart of the EV — its battery — India must **replicate** its approach to localising ICE manufacturing, which is to **mandate** joint ventures with local ICE or EV makers, and gradually allow for a complete open market.

[Practice Exercise]

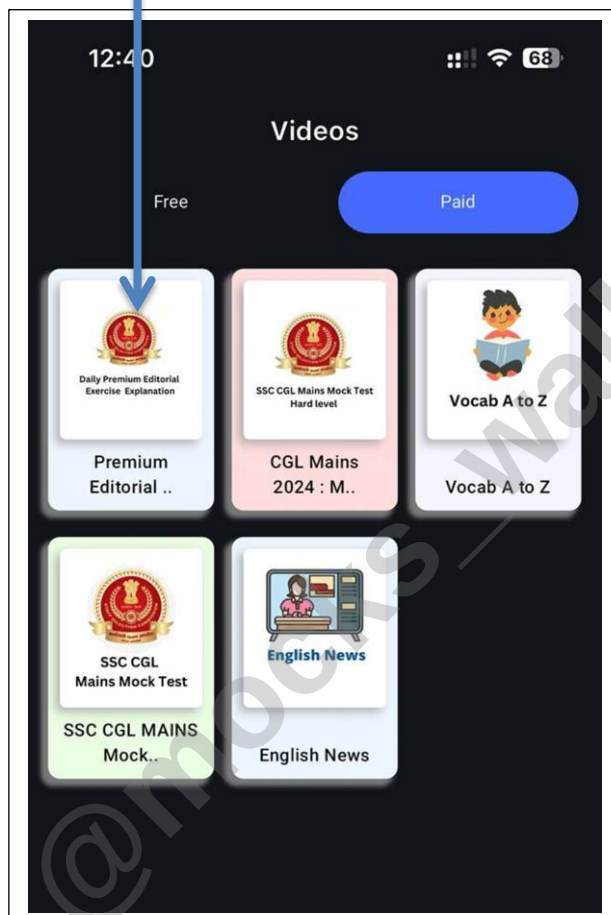
- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

Vocabulary

1. **Take a turn** (phrase) – change direction, shift course, alter path, deviate, veer मोड़ लेना / दिशा बदलना
2. **Concessional** (adjective) – discounted, reduced, preferential, subsidised, lower रियायती
3. **Import duty** (noun) – taxes imposed by a government on goods imported into a country आयात शुल्क
4. **Contingent** (on) (adjective) – dependent, conditional, subject to, based on, reliant निर्भर
5. **Localise** (verb) – adapt, regionalise, domesticate, indigenise, internalise स्थानीय बनाना
6. **Add** (to) (verb) – increase, enhance, contribute, supplement, augment बढ़ाना
7. **Bouquet** (noun) – range, collection, array, assortment, cluster विविधता / समूह
8. **Fall short of** (phrase) – fail to meet, underperform, lack, miss, not achieve कमी रह जाना
9. **Address** (verb) – tackle, deal with, respond to, handle, confront समाधान करना
10. **Pressing** (adjective) – urgent, critical, serious, immediate, crucial अत्यावश्यक
11. **Decarbonise** (verb) – reduce carbon emissions, cut pollution, depollute, neutralise emissions, go green कार्बन उत्सर्जन कम करना
12. **Mobility** (noun) – transportation, movement, transit, conveyance, locomotion गतिशीलता / आवाजाही
13. **Economies** (noun) – countries in terms of GDP अर्थव्यवस्थाएँ
14. **Outlay** (noun) – expenditure, investment, spending, allocation, cost व्यय / खर्च
15. **Expand** (verb) – increase, grow, extend, widen, amplify विस्तार करना
16. **Ambitious** (adjective) – bold, determined, enterprising, visionary, daring महत्वाकांक्षी
17. **Couple** (with) (verb) – combine, link, connect, associate, integrate जोड़ना / साथ लाना
18. **Joint venture** (noun) – a business entity created by two or more parties, generally characterized by shared ownership, shared returns and risks, and shared governance संयुक्त उद्यम
19. **Cumulative** (adjective) – total, combined, collective, aggregate, accumulated संचयी
20. **Incentive** (noun) – motivation, encouragement, stimulus, boost, inducement प्रोत्साहन

21. **Rapid** (adjective) – fast, quick, swift, speedy, accelerated तीव्र
22. **Infrastructure** (noun) – framework, foundation, base, facilities, system आधार संरचना
23. **Deployment** (noun) – implementation, use, positioning, operation, utilisation कार्यान्वयन
24. **Adoption** (noun) – acceptance, implementation, use, embrace, uptake अंगीकरण
25. **Account for** (phrasal verb) – represent, make up, contribute, comprise, constitute बनाना
26. **Remainder** (noun) – rest, balance, residue, leftover, others बाकी हिस्सा
27. **Vertical integration** (noun) – the combination in one firm of two or more stages of production normally operated by separate firms.
28. **Aid** (verb) – help, assist, support, facilitate, back मदद करना
29. **Economies of scale** (noun) – a proportionate saving in costs gained by an increased level of production.
30. **Conventional** (adjective) – traditional, standard, regular, orthodox, established पारंपरिक
31. **Repurpose** (verb) – modify, adapt, redesign, retool, convert नया उपयोग देना
32. **Layer** (verb) – a division or section within the software architecture, often related to specific functionalities or responsibilities
33. **Software-as-a-service** (noun) – It means accessing software through the internet without downloads
34. **Replicate** (verb) – copy, reproduce, imitate, duplicate, mimic दोहराना / नकल करना
35. **Mandate** (verb) – require, order, enforce, command, direct आदेश देना / अनिवार्य करना

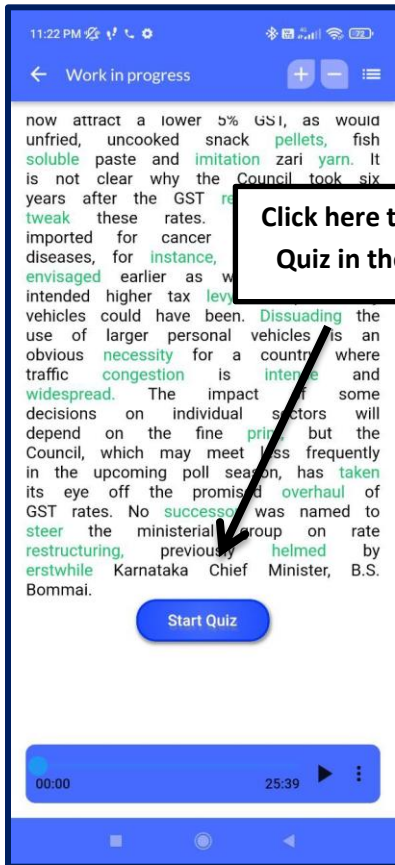
Video solution of Practice Exercise from this Editorial is exclusively available on App.



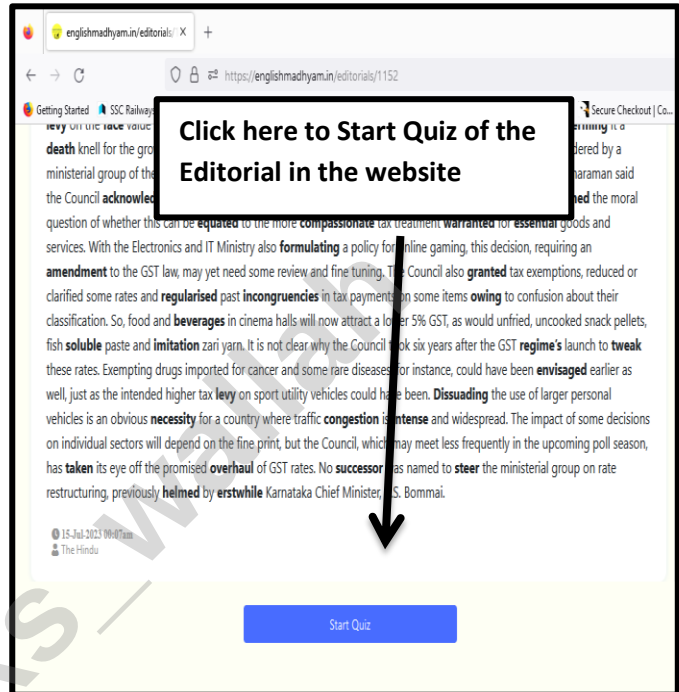
Summary of the Editorial

1. **Policy Move:** On June 2, India introduced a concessional 15% import duty on fully built EV units, aiming to boost domestic manufacturing.
2. **Investment Requirement:** EV manufacturers must invest at least ₹4,150 crore in three years and achieve 25% domestic value addition (DVA), increasing to 50% in five years.
3. **Import Cap:** Under the SPMEPCI scheme, each manufacturer is allowed to import a maximum of 8,000 fully built units annually for five years.
4. **Policy Objective:** The new scheme seeks to promote EV adoption and manufacturing through a mix of incentives and localisation requirements.
5. **Key Challenge:** Despite the new policies, India is still lacking in one critical area — **technology transfer**, especially in EV battery tech.
6. **Late Start:** India began its EV journey in 2015, lagging behind other major economies like China and the U.S.
7. **FAME Scheme:** India's FAME initiative started with ₹895 crore, expanded to ₹10,000 crore in 2019 to push hybrid and electric vehicles.
8. **China's Strategy:** China launched its EV subsidy program in 2009, required joint ventures for foreign EV makers, and ensured large-scale **technology transfer**.
9. **Massive Investment:** China offered around \$230 billion in cumulative incentives over 15 years, the largest globally.
10. **Result of China's Policy:** China became the world leader in EV production and consumption, with 11.3 million of 17 million global EV sales in 2024.
11. **Charging Infrastructure:** China also rapidly developed charging networks, further accelerating EV adoption.
12. **U.S. Efforts:** The U.S. started its push in 2010, expanded it through the Inflation Reduction Act, but lags behind China in adoption rates.
13. **Battery Supply Chain:** China's vertical integration in battery supply chains gave it scale and pricing advantages.
14. **India's Current Plan:** India's 25% DVA target will initially depend on repurposed ICE components and software overlays.
15. **Way Forward:** For deeper localisation and technological independence, India must mandate joint ventures with local EV/ICE players, similar to how it built its ICE manufacturing sector.

Practice Exercise: SSC Pattern Based



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